Phase 2 | Requirement Analysis

Data Flow Diagram

AI-Powered Travel Planner | Smart Situational Itinerary Engine

Live URL:	https://travel-planner-iota-plum.vercel.
GitHub:	github.com/itskian2007/travel-planner
Tech:	React + FastAPI + PostgreSQL + Vercel
Team:	itskian2007

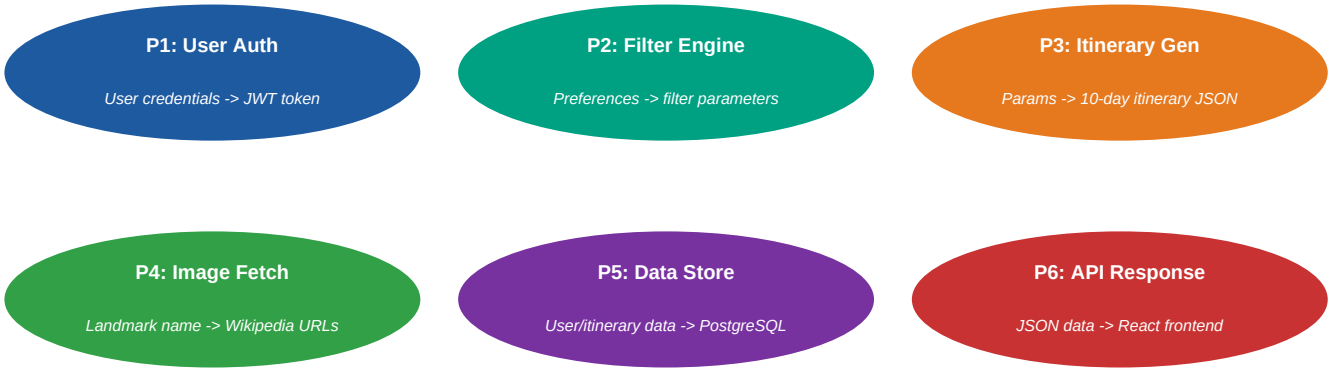
1. System Overview

The Travel Planner system follows a 3-tier architecture: React frontend, FastAPI backend, and PostgreSQL database, with an external Wikipedia API for image data. Data flows through the system via REST API calls with JSON payloads.

2. Level 0 - Context Diagram

3. Level 1 - System DFD Processes

Decomposed data flows through the main system processes:



4. Data Stores

Store ID	Table Name	Key Fields
DS1	Users Table	user_id, name, email, password_hash, created_at
DS2	Destinations	dest_id, name, country, description, lat, lng, image_urls
DS3	Itineraries	itinerary_id, user_id, dest_id, mood, budget, days_json
DS4	Preferences	pref_id, user_id, diet, persona, accessibility, tier
DS5	Image Cache	landmark_name, wikipedia_url, cached_at

5. API Endpoints Data Flow

Method	Endpoint	Input Data	Output Data	Module
POST	/api/auth/register	User JSON body	JWT + user_id	Auth Module
POST	/api/auth/login	email + password	JWT token	Auth Module
GET	/api/destinations	JWT header	destinations[]	Dest Module
POST	/api/itinerary/generate	prefs JSON	itinerary JSON	Itinerary Engine
GET	/api/images/{landmark}	landmark_name	image_url[]	Wikipedia API
GET	/api/destinations/{id}	dest_id	full destination details	Dest Module